Assignment 1 (Individual Assignment) (30%)

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Instructions

Assignment 1 — Individual (30% of course)

Due: End of Week 6
Dataset: Eight data tables (DimDate, DimDiagnosis, DimFacility, DimPatient, DimPayer, DimProcedure, DimProvider, FactEncounter) are provided.
Purpose: Consolidate Modules 1–5: Power Query ETL, data profiling, data model design, basic DAX and visualization.

Learning outcomes

- Present and evaluate authentic examples of Power BI solutions in descriptive, predictive, and prescriptive analytics.
- Get data from different data sources, profile, clean, transform, and load the data into a data warehouse.
- Design and develop a data model, create measures by using DAX, and optimize the model performance.
- Create reports and dashboards to visualize data, enhance reports to expose insights, and perform advanced analysis for AI insights.

Tasks (required)

1. **Power Query / ETL**
    - Import all the data tables (DimDate, DimDiagnosis, DimFacility, DimPatient, DimPayer, DimProcedure, DimProvider, FactEncounter) into Power BI.
    - Rename all the tables for readability. For example:

      DimDate → Calendar Table

      DimPatient → Patient Table

      DimFacility → Facility Table

    - Inspect all data tables and perform comprehensive data profiling data (missing values, outliers, empty string, uninterpretable characters, inconsistent types).
    - Add the following fields in the “DimDate”.
        a. Start of the Year
        b. Start of the Quarter
        c. Start of the Month

    d. Month Name
    e. Week of the Year
  - In the Patient Table (DimPatient), create a new column called Patient Profile by combining columns Gender, AgeGroup and State using a comma (,) as a separator.
  - Use the **ReadmissionFlag** (currently `0` or `1`) to create a **Readmission Category** column:

    Yes if 1, No if 0
  - Document each transformation step in Power Query.

1. **Data model**
  - From the properties pane, mark the primary keys of the dimension tables and build a data model based on the Star schema.
  - Select correct data types of the fields in each model
  - In Report View, add a Matrix Visual consists of fields from every table to show that the model is build correctly.
2. **DAX measures (minimum 10)**
  - Create at least 10 DAX measures. Make sure that there must be at least one from Math and stat function, counting function, Logical and conditional function, Calculate, and Time Intelligence
4. **Visualization**: Use different **report layouts** to explore relationships between parameters. Apply descriptive analysis with a variety of visuals such as bar charts, pie charts, line charts, Matrix and tables and add interactive slicers for filtering.
5. **Deliverable report (3–4 pages) and Video presentation (3- 5 minutes)**
  - Brief dataset description; summary of ETL steps.
  - Diagram and explanation of model relationships.
  - List of DAX measures with one-line explanation of purpose & logic.
  - 3–5 short actionable insights (supported by visuals/screenshots).
  - Brief note on limitations and one optimization you applied.

Deliverables

- .pbix file: lastname_firstname_Assn1.pbix
- PDF/Word report: lastname_firstname_Assn1.pdf or .docx
- Video clip of 3-5 minutes
- Submit to LMS by end of week 5.